SOLUTIONS OF DAY-3 QUESTIONS

Question 1:
https://codeforces.com/problemset/problem/411/A
Solution:

```
#include<iostream>
#include<bits/stdc++.h>

using namespace std;

int main()
{
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
    // there are 4 conditions here
    // 1) at least 5 characters
    // 2) at least one large letter
    // 3) at least one small letter
    // 4) at least one digit
    // we are going to solve this using typecasting and ascii values of
characters
    // if we typecast a character into integer then it would return ascii
value of that character
    // ascii values of digits('0' to '9') would be in the range (48 to 57)
    // ascii values of large letters('A' to 'Z') would be in the range (65
to 90)
    // ascii values of small letters('a' to 'z') would be in the range (97
to 122)
    string s;
    cin>>s;
    int n=s.length();
    bool a=false,b=false,c=false,d=false;
    // 4 conditions => 4 bool variables
    if(n>=5) a=true;
    int x;
    for(int i=0;i<n;i++)
    {
        x=(int)(s[i]);
        // typecasted character s[i] into integer
        if((x>=48) && (x<=57)) b=true; // for digits
```

```
        if((x>=65) && (x<=90)) c=true; // for large letter
        if((x>=97) && (x<=122)) d=true; // for small letter
    }
    // if 4 conditions are met then we would print Correct
    if(a && b && c && d)
    {
        cout<<"Correct"<<endl;
    }
    else
    {
        cout<<"Too weak"<<endl;
    }
    // n-> length of string
    // TC = O(n)
    // SC = O(1)(generally we don't consider input in space complexity,we
only consider extra space used)
    // key learnings : about typecasting and ascii values of different
characters
    // make sure to read about ascii values of different characters
}
```

Question 2:
https://codeforces.com/problemset/problem/26/A
Solution:

```
#include<iostream>
#include<bits/stdc++.h>
using namespace std;

bool isPrime(int n)
{
    int factors=0;
    for(int i=1;i<=n;i++)
    {
        if(n%i==0) factors++;
    }
    if(factors==2) return true;
    return false;
}
```

```
int main()
{
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
    // we want number of almost primes till n
    // firstly we know that a prime number has exactly 2 divisors(1 and
itself)
    // we have a boolean function that tells whether the number is prime
or not
    // from 1 to n we have stored whether it is prime or not in an array
    // we check for each i from 1 to n whether it is almost prime or not
by
    // checking if it has exactly 2 prime factors
    int n;
    cin>>n;
    bool v[n+1];
    v[0]=0;
    for(int i=1;i<=n;i++)
    {
        if(isPrime(i)) v[i]=1;
        else v[i]=0;
    }
    int ans=0;
    for(int i=1;i<=n;i++)
    {
        int temp=0;
        for(int j=1;j<=i;j++)
        {
            if(i%j==0 && v[j]) temp++; // we only increment temp if j is a
prime and it is a divisor of i
        }
        if(temp==2) ans++; // checks whether it has exactly 2 prime
divisors
    }
    cout<<ans<<endl;
    // TC=O(n^2) and SC=O(n) (array v)
    // key learnings : about how to tell whether a number is prime or not
    // and also we can optimize isPrime function by a little bit
    // Hint : iterate from 1 to root n instead of 1 to n
}
```

Question 3:
https://codeforces.com/problemset/problem/152/A
Solution:

```
#include<iostream>
#include<bits/stdc++.h>

using namespace std;

int main()
{
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
    // see the constraints(1<=n,m<=100)
    // they will be giving us a maximum of 100 digit input and we cant
store them in either int or longlong
    // int can store upto 10^9 and longlong can store upto 10^18
    // thatswhy we use string and store the input in it
    int n,m;
    cin>>n>>m;
    vector<string> v(n);
    for(int i=0;i<n;i++)
    {
        cin>>v[i];
    }
    // we are creating a vector in which we store maximum marks for each
subject
    vector<int> maxi(m);
    for(int i=0;i<m;i++)
    {
        int x=-1;
        // initialized x with -1(can be initialized with any value less
than or equal to 1)
        for(int j=0;j<n;j++)
        {
            // here note that v[j][i] is a character and not an integer
and we are subtracting with '0'
            // if v[j][i] is '1' then v[j][i]-'0' would return us 1
            // we are again using the ascii value concept here
```

```
            // subtracting the ascii value of '0' from the ascii value of
character v[j][i]
            int y=v[j][i]-'0';
            x=max(x,y);
            // max(a,b) would return the largest among those 2 parameters
        }
        maxi[i]=x;
        // we stored the maximum marks of ith subject in maxi[i]
    }
    int ans=0;
    for(int i=0;i<n;i++)
    {
        bool succesful=false;
        for(int j=0;j<m;j++)
        {
            // for ith student we are checking whether he/she is
successful in jth subject
            int x=v[i][j]-'0';
            // again note that v[i][j] is a character here
            // if maxi[j] matches with x then that means that he is
successful
            if(x==maxi[j])
            {
                succesful=true;
                break;
                // break statement simply breaks from the for loop
            }
        }
        if(succesful) ans++;
    }
    cout<<ans<<endl;
    // TC = O(n*m) SC = O(m)(maxi vector)
    // key concepts : choosing the correct datatype to store the given
input and its constraints
}
```